

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

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# Geometry and Trigonometry

# 07

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

**Q1****70**

The correct answer is 70. Based on the figure, the angle with measure  $110^\circ$  and the angle vertical to the angle with measure  $x^\circ$  are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure  $x^\circ$  also has measure  $x^\circ$ . It's given that lines  $s$  and  $t$  are parallel. Therefore, same side interior angles between lines  $s$  and  $t$  are supplementary. It follows that  $x + 110 = 180$ . Subtracting 110 from both sides of this equation yields  $x = 70$ .

**Q2****A****A. 9**

Choice A is correct. The area  $A$ , in square inches, of a rectangle is the product of its length  $l$ , in inches, and its width  $w$ , in inches; thus,  $A = lw$ . It's given that the area of a rectangle is 630 square inches and the length of the rectangle is 70 inches. Substituting 630 for  $A$  and 70 for  $l$  in the equation  $A = lw$  yields  $630 = 70w$ . Dividing both sides of this equation by 70 yields  $9 = w$ . Therefore, the width, in inches, of this rectangle is 9.

Choice B is incorrect. This is the length, not the width, in inches, of the rectangle.

Choice C is incorrect. This is half the area, in square inches, not the width, in inches, of the rectangle.

Choice D is incorrect. This is the difference between the area, in square inches, and the length, in inches, of the rectangle, not the width, in inches, of the rectangle.

**Q3****B****B.** 77

Choice B is correct. The triangle shown is a right triangle, where the interior angle shown with a right angle symbol has a measure of  $90^\circ$ . It's shown that the other two interior angles measure  $13^\circ$  and  $a^\circ$ . The sum of the measures of the interior angles of a triangle is  $180^\circ$ ; therefore,  $90 + 13 + a = 180$ . Combining like terms on the left-hand side of this equation yields  $103 + a = 180$ . Subtracting 103 from both sides of this equation yields  $a = 77$ .

Choice A is incorrect. This is the measure, in degrees, of the other acute interior angle of the right triangle, not the value of  $a$ .

Choice C is incorrect. This is the measure, in degrees, of the right angle of the right triangle, not the value of  $a$ .

Choice D is incorrect. This is the sum of the measures, in degrees, of the other two interior angles of the right triangle, not the value of  $a$ .

**Q4****A****A.** 8

Choice A is correct. It's given that the area of a square is 64 square inches. The area  $A$ , in square inches, of a square is given by the formula  $A = s^2$ , where  $s$  is the side length, in inches, of the square. Substituting 64 for  $A$  in this formula yields  $64 = s^2$ . Taking the positive square root of both sides of this equation yields  $8 = s$ . Thus, the side length, in inches, of this square is 8.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the area, in square inches, of the square, not the side length, in inches, of the square.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q5****A****A.** 52

Choice A is correct. The sum of the measures of the three interior angles of a triangle is  $180^\circ$ . Therefore,  $31 + 2b + a = 180$ . Since it's given that  $a = 45$ , it follows that  $31 + 2b + 45 = 180$ , or  $2b = 104$ . Dividing both sides of this equation by 2 yields  $b = 52$ .

Choice B is incorrect and may result from a calculation error. Choice C is incorrect. This is the value of  $a + 31$ . Choice D is incorrect. This is the value of  $2b$ .

**Q6****C****C.** 42

Choice C is correct. The area,  $A$ , of a rectangle is given by the formula  $A = lw$ , where  $l$  represents the length of the rectangle and  $w$  represents its width. It's given that the rectangle has a length of 7 inches and a width of 6 inches. Substituting 7 for  $l$  and 6 for  $w$  in the formula  $A = lw$  yields  $A = 76$ , or  $A = 42$ . Thus, the area, in square inches, of the rectangle is 42.

Choice A is incorrect. This is the sum, not the product, of the length and width of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is twice the area, in square inches, of the rectangle.

**Q7****D****D.** 40

Choice D is correct. Since  $\angle BCD$  and  $\angle BCA$  form a linear pair of angles, their measures sum to  $180^\circ$ . It's given that the measure of  $\angle BCD$  is  $110^\circ$ . Therefore,  $110^\circ + \angle BCA = 180^\circ$ .

Subtracting  $110^\circ$  from both sides of this equation gives the measure of  $\angle BCA$  as  $70^\circ$ . It's also given that the measure of  $\angle BAC$  is equal to the measure of  $\angle BCA$ . Thus, the measure of  $\angle BAC$  is also  $70^\circ$ . The measures of the interior angles of a triangle sum to  $180^\circ$ . Thus,  $70^\circ + 70^\circ + x^\circ = 180^\circ$ . Combining like terms on the left-hand side of this equation yields  $140^\circ + x^\circ = 180^\circ$ . Subtracting  $140^\circ$  from both sides of this equation yields  $x^\circ = 40^\circ$ , or  $x = 40$ .

Choice A is incorrect. This is the value of the measure of  $\angle BCD$ . Choice B is incorrect. This is the value of the measure of each of the other two interior angles,  $\angle BCA$  and  $\angle BAC$ . Choice C is incorrect and may result from an error made when identifying the relationship between the exterior angle of a triangle and the interior angles of the triangle.

**Q8****A****A.** 18

Choice A is correct. The volume,  $V$ , of a rectangular prism is given by the formula  $V = lwh$ , where  $l$  is the length of the base,  $w$  is the width of the base, and  $h$  is the height of the prism. It's given that each base of a right rectangular prism has a length of 19 inches and a width of 8 inches, and that the prism has a volume of 2,736 cubic inches. Substituting 19 for  $l$ , 8 for  $w$ , and 2,736 for  $V$  in the formula  $V = lwh$  gives  $2,736 = (19)(8)(h)$ , or  $2,736 = 152h$ . Dividing each side of this equation by 152 yields  $18 = h$ . Therefore, the height, in inches, of the prism is 18.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.

**Q9****C****C.**  $91^\circ$ 

Choice C is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is  $180^\circ$ . It's given that in  $\triangle XYZ$ , the measure of  $\angle X$  is  $23^\circ$  and the measure of  $\angle Y$  is  $66^\circ$ . It follows that the measure of  $\angle Z$  is  $180 - 23 - 66^\circ$ , or  $91^\circ$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the sum of the measures of  $\angle X$  and  $\angle Y$ , not the measure of  $\angle Z$ .

Choice D is incorrect and may result from conceptual or calculation errors.

**Q10****C****C.** 18

Choice C is correct. It's given that triangle  $TUV$  is similar to triangle  $XYZ$ , and  $\overline{TU}$  corresponds to  $\overline{XY}$ . If two triangles are similar, then the ratio of their perimeters is equal to the ratio of their corresponding sides. It's given that the perimeter of triangle  $TUV$  is 50, the perimeter of triangle  $XYZ$  is 150, and the length of  $\overline{TU}$  is 6. Let  $n$  represent the length of  $\overline{XY}$ . It follows that  $\frac{50}{150} = \frac{6}{n}$ , or  $\frac{1}{3} = \frac{6}{n}$ . Multiplying each side of this equation by  $n$  yields  $\frac{n}{3} = 6$ . Multiplying each side of this equation by 3 yields  $n = 18$ . Therefore, the length of  $\overline{XY}$  is 18.

Choice A is incorrect. This is the solution to  $\frac{3}{1} = \frac{6}{n}$ , not  $\frac{1}{3} = \frac{6}{n}$ .

Choice B is incorrect. This is the length of  $\overline{TU}$ , not  $\overline{XY}$ .

Choice D is incorrect. This is the sum of the length of  $\overline{TU}$  and the perimeter of triangle  $TUV$ , not the length of  $\overline{XY}$ .

**Notes**

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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